User Lifecycle Management - Admin Console

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What is User Lifecycle Management?

User lifecycle management is a comprehensive process that involves managing a user's identity and access throughout their association with an organization.

The User Lifecycle Management process outlines the steps involved in user registration, access provisioning, access deprovisioning, and governance. It serves as a reference for admins, super admins and other business personas to efficiently manage user roles and permissions.

User Lifecycle Management is managed by different user personas, each with specific permission:

Organization owner

Organization owners are responsible for managing organizational-level configurations. Whichever application is a part of that organization, will have those groups also applied to it. Organization owners are on a higher level than Applications Owner. Organization owners can create organization level groups, add or remove applications from the organizations which App Owners or App admins are unable to do. Organization Owners have full access

to all applications and users. They have the ability to create, modify, and remove users and roles. Organization owners can manage policies and governance.

Application Owner

An application owner is responsible for a specific application. They can manage users and access within that application. They are assigned by an application admin and can also assign others as application owners. Application owners can create groups, assign users, and define roles. However, they cannot view other applications.

Group Owner

A group owner has restricted access within the application system. They can manage user access within a specific group but cannot change all configurations. Group owners can manage users within their assigned group or division, but they cannot access users outside their division.

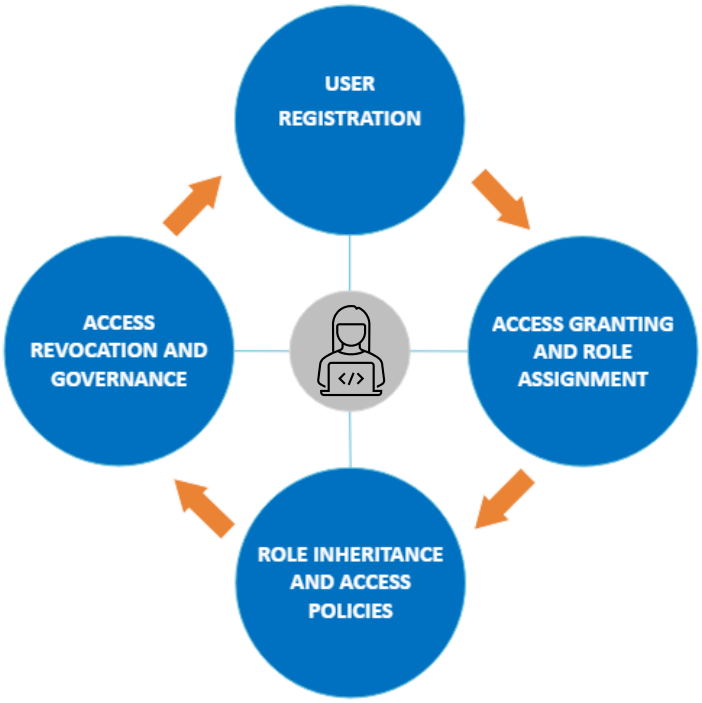
User Administrator

A user administrator manages user-related operations, such as user registration, adding or removing users, assigning roles, and handling security concerns like deactivating compromised accounts, ensures users have the correct permissions based on governance policies. They can search for users via email or username and manage their access within the system.

Overview of Efficient User Management

This feature in the Admin Console will enable you to invite new users to join your team. You can explore the features that enable you to view assigned role and permissions to the process of managing user identities, roles, and access rights throughout their engagement with the organization's system. It will ensure secure, role-based access while maintaining compliance with policies.

User Lifecycle Stages

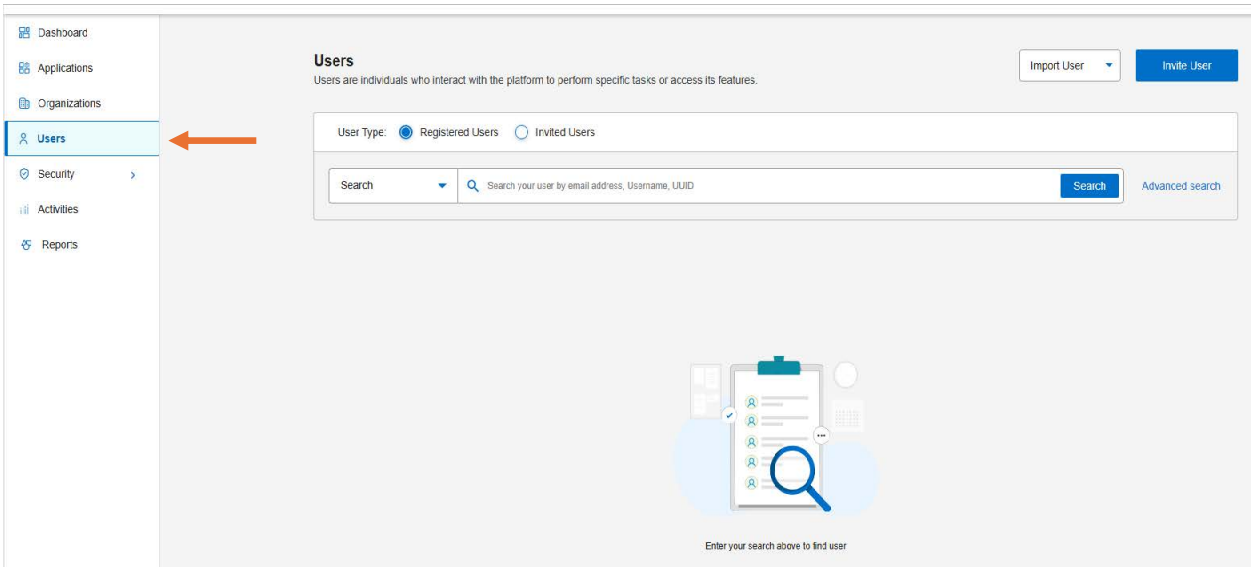


The lifecycle of a user consists of the following stages:

1. User Registration
2. Access Granting and Role Assignment
3. Compliance and Access Policies
4. Access Revocation and Governance

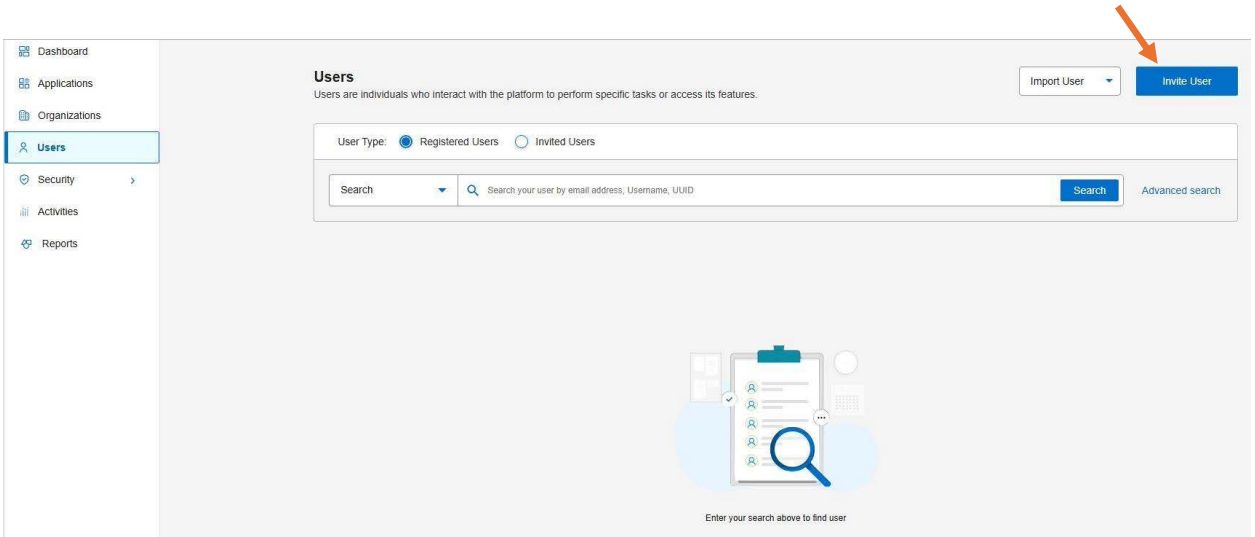
User Registration

This tab is used for onboarding new users. User invitation is the process by which an individual creates an account within an application or system. This typically involves providing personal information, such as an email address and password, and completing any required verification steps. Once registered, the user gains access to the application's features and functionalities (only when the user is invited through the permission **Assign access to this invitation**).



Steps to Invite a User and thereafter Registering the User

1. Locate and click **Invite User** within the application.

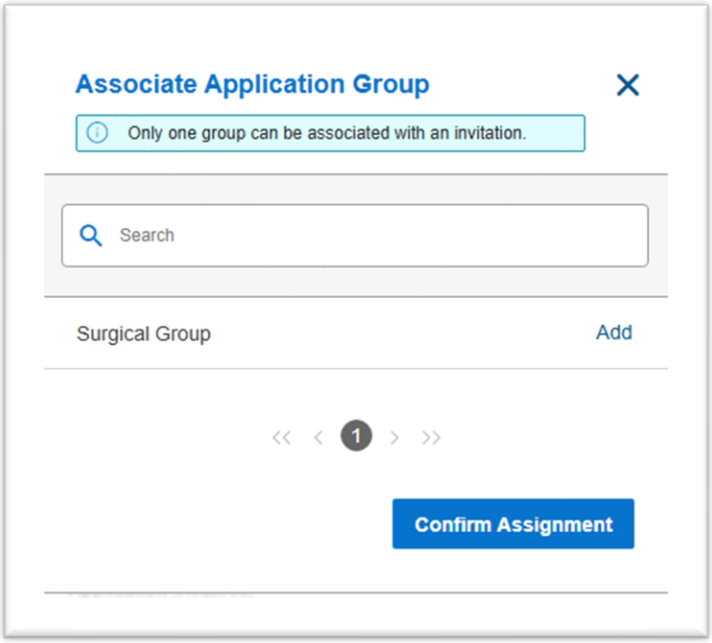


2. Enter the user's first and last name, email address, and username.
3. Select the existing application from the dropdown to grant access to the invited user.
4. Choose the **Application Redirect URL** from the dropdown.
   For example, email address: ssmith@guerrillamail.com, name: Steve Smith, username: Ssmith

5. Click **Assign access** to this invitation.

Steps to follow in the Associate Application Group

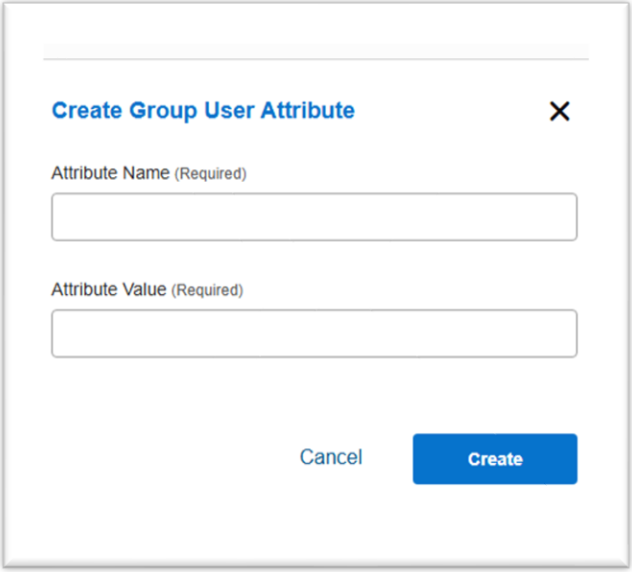
1. Click **Associate Application Group**.
2. Search the group name, and click **Add**.
3. Click **Confirm Assignment**.



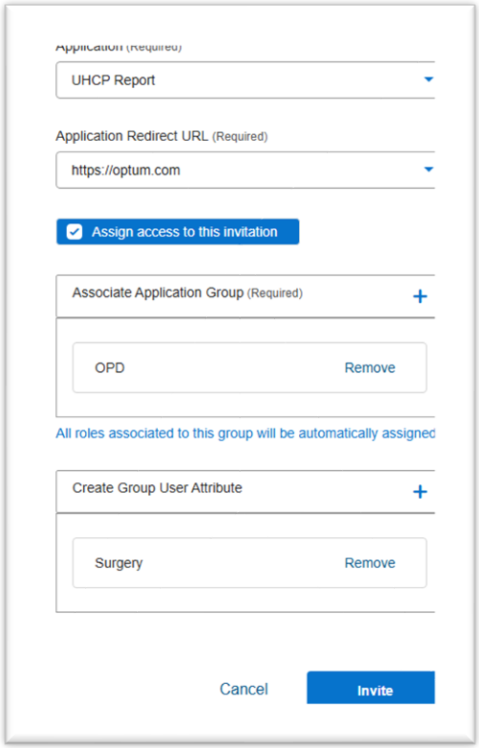
Notes:

- Attribute Name is a user-defined identifier (key) for additional metadata attached to a user's access profile.
- Attribute Value is the specific data assigned to the attribute name, used for policy enforcement or decision-making.
  For example:
  Attribute Name: Role
  Attribute Value: Healthcare Admin
  OR
  Attribute Name: Persona
  Attribute Value: Super Administrator

4. Create **Group User Attribute**
5. Fill **attribute name**
6. Fill **attribute value**
7. Click **Create.**



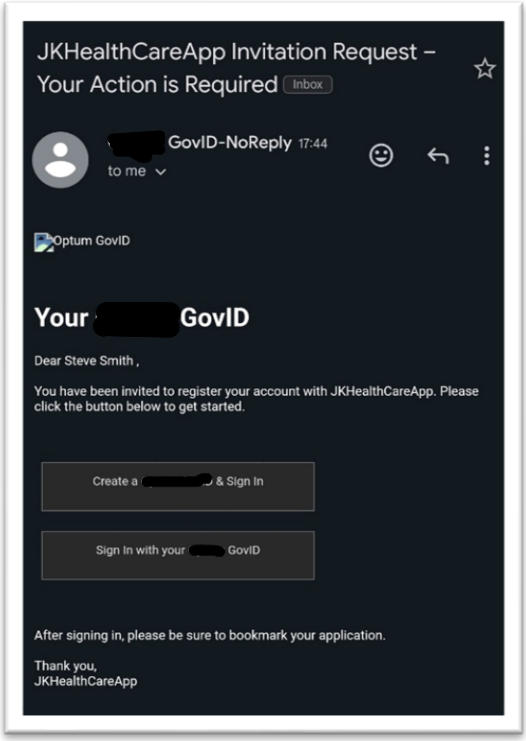
8. Click the **Invite** button. The user will receive an email with a hyperlink to register.



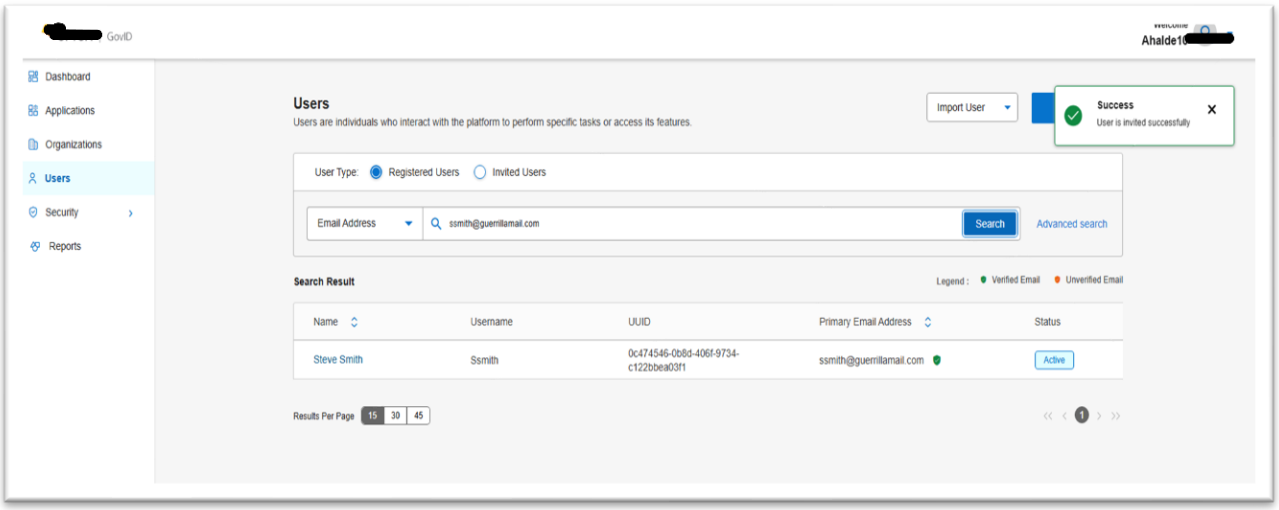
9. The invited user (e.g., Steve Smith) will receive an email containing a hyperlink to log in or sign in with their XXXX Gov ID.
10. Users can self-register via a portal or through the invitation received.

1. The user clicks on the hyperlink and follows the registration flow to complete their registration.

2. Upon registration, users receive an email confirmation and authentication link.



3. Once registered, the user will appear in the **Registered Users** type under the **User** section.



Access Granting and Role Assignment

Role Assignment is the process of linking predefined roles (bundles of permissions) to users or groups. Roles define:

- What a user can access (e.g., applications, data).
- How they can interact (e.g., view, edit, approve).

The two types Role Assignment are:

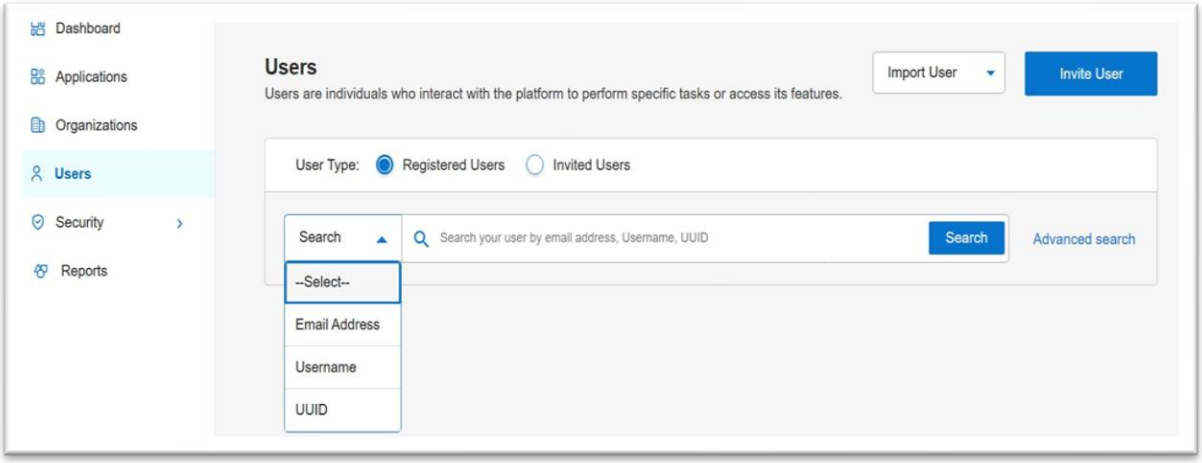
- User based role assignment
- Group based role assignment

User based Role Assignment

This step involves assigning the attribute necessary permissions to users. Users are assigned to roles and groups based on their department or application.

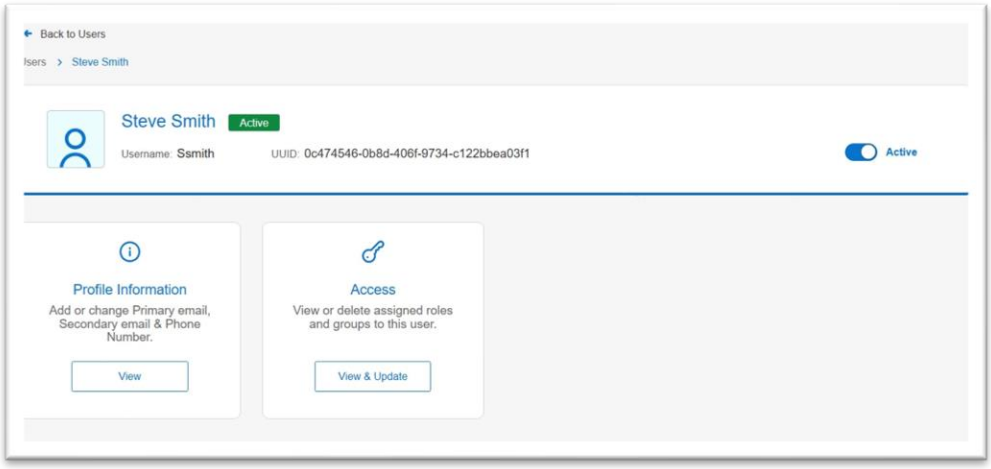
Steps

1. Navigate to the **Users** section.
2. Select **Registered Users** in User Type.
3. Select **Email Address / Username / UUID** from the **Search** dropdown.

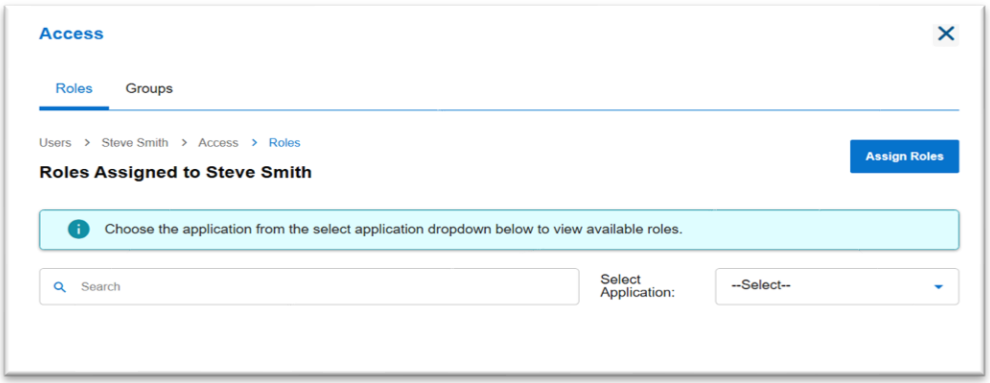


4. Type **Email Address / Username / UUID** of the user as per the selection.
   For example, ssmith@guerrillamail.com / Ssmith / 0c474546-0b8d-406f-9734-c122bbea03f1
5. Click **Search**
6. Click on the name of the user.

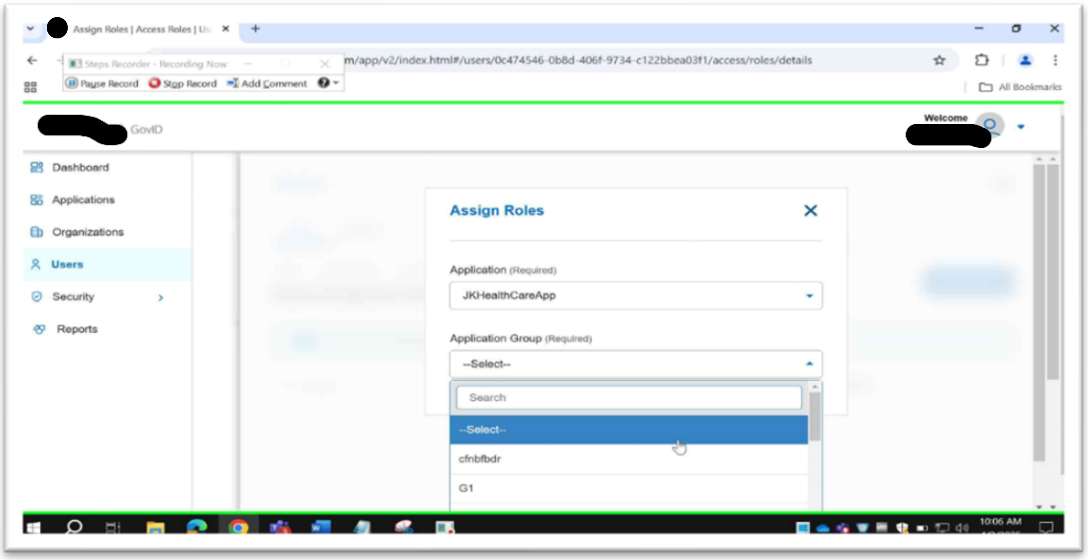
7. **Access** > **Assign Roles**



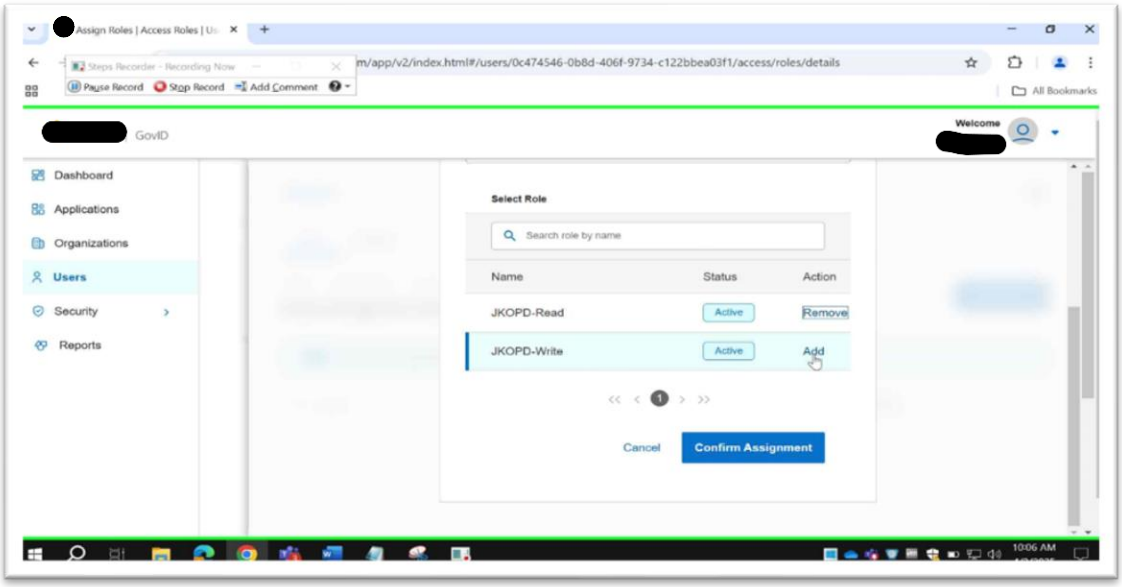
8. Select an **application** from the dropdown



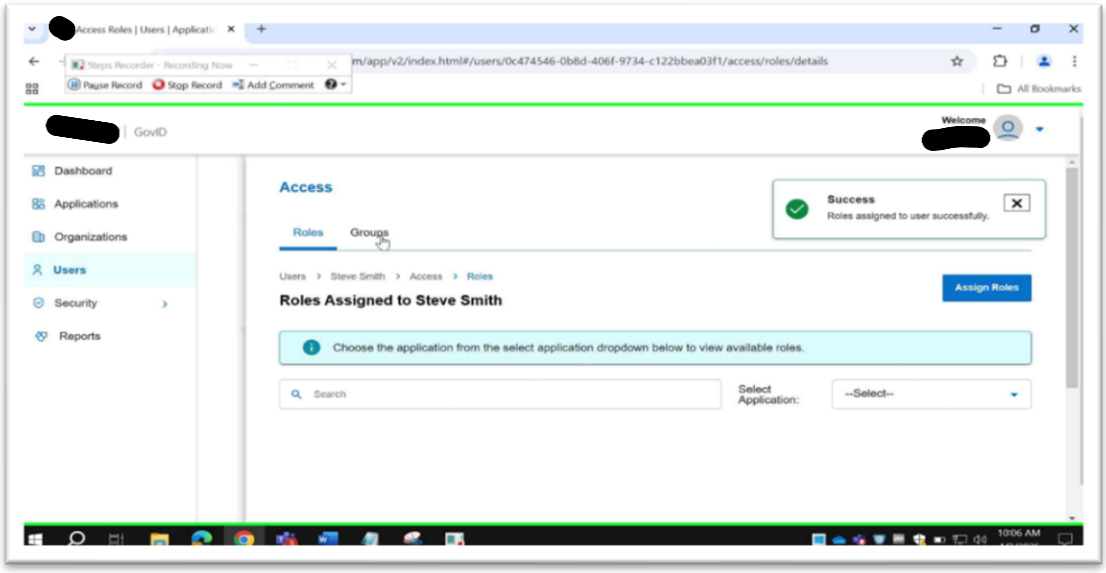
9. Select **application group** from the dropdown



10. **Add** > **Confirm Assignment**



11. Users will successfully be assigned with roles.



Group Based Role Assignment

Group-based Role Assignment automatically grants access permissions to users by assigning them to predefined groups (e.g., departments, teams, or divisions). Roles are attached at the group level instead of individually.

Steps:

1. Go to the **Access** section within the application to view the user's current access.

2. If no access is displayed, select the application for which you want to view or assign roles.
   Example: Type and select **JK Healthcare** to view roles for that application.
3. The system will display the roles and groups assigned to the user for the selected application.
   Example: The user might be an admin for "JK Healthcare".
4. Assign New Roles
   a) To assign new roles, click on **Assign Roles**.
   b) Select the application for which you want to assign roles to the user.
   c) Choose the appropriate group within the application.
      Group-based access ensures users inherit permissions automatically.
      Example: Select a group with new operating coordinates.
   d) Click **Add** to assign the selected roles and groups to the user.
   e) Confirm the assignment to finalize the process.
5. Verify Role Assignment
   a) Search for the user to verify that the new roles have been assigned.
      Example: The assigned role will be visible in the user's profile.
   b) If the user is a member of a group, this information will be displayed in the **Group** tab.
      Example: The group that the role belongs to will be listed.

6. Advanced Search Option

   - If you don't have the user's email address, username, or OHID, use the advanced search option.
   - Search using the first name and last name.
     Example: If user's name is Steve Smith, then enter the first name Steve and last name Smith to find the user.

7. Bulk Import for Access Granting
   1. Use the **Import User** functionality for bulk access granting.
   2. Prepare an Excel file with user details in the predefined format.
      Example: The file should include columns for OHID, username, and roles.
   3. Upload the file to import users and assign access.
      - New users will receive an email invitation.
      - Existing user s will have access assigned directly.

Fast-Track User Registration with Access

Ensures users are productive immediately after registration by granting access during the onboarding process.

Steps:

1. An admin (e.g., application/group owner) invites a user via the Admin Console, optionally assigning access (roles/groups) upfront.
2. The user receives an email with a registration link. Upon clicking, they:
   a) Create an Xxxx/One Healthcare ID (if new) or sign in (if existing).
   b) Complete registration (e.g., set password, security questions).
   c) Are automatically granted pre-assigned access and redirected to the application (e.g., Xxxx RX).

   Note: No separate access-grant step is needed; roles/groups are inherited instantly.

   Use Case: Ideal for providers/customers who need immediate access (e.g., healthcare providers joining a portal).

Direct Self-Registration Process

Purpose: Allows users to register independently but requires manual access assignment afterward.

Process:

1. A user visits an application (e.g., Xxxx RX) and clicks **Register with One Healthcare ID**.
2. They fill out a registration form and are redirected back to the application post-signup.
3. No access is granted automatically; admins must manually assign roles/groups later via Admin Console. Users register directly via a portal and receive access after authentication.

Invitation-Based Registration

- Admins can invite users via email.
- Upon clicking the invitation link, users are automatically assigned to a group.
- Bulk User Import and Access Assignment
- Large-scale user onboarding is supported through CSV imports.
- Admins define roles and groups in advance.

	DIRECT SELF REGISTRATION	INVITATION BASED REGISTRATION	BULK USER IMPORT + ACCESS ASSIGNMENT
Initiated By	User (via application portal	Admin (sends email invite) **FAST-TRACK REGISTRATION** Admin (via Admin Console)	Admin (uploads CSV file)
Access Assignment	Manual post-registration by admin	Optional (can assign during invite) **FAST-TRACK REGISTRATION** Pre-assigned during invitation (roles/groups)	Pre-assigned in bulk via CSV
User Steps	Self-register → Wait for access	Click invite link → Register **FAST-TRACK REGISTRATION** Click invite link → Register → Access granted	Receive invite (if new) → Register
Bulk Support	No (individual only)	Limited (single-user invites) **FAST-TRACK REGISTRATION** Yes (via CSV import)	Yes (supports numerous users)
Use Case	Public user onboarding (e.g., members)	Controlled onboarding (e.g., contractors) **FAST-TRACK REGISTRATION** Critical roles needing instant access (e.g., providers)	Mass onboarding (e.g., enterprise teams)
Example	Member signs up via Xxxx portal	Contractor invited with temporary access **FAST-TRACK REGISTRATION** Provider invited to Xxxx RX with pre-set roles	HR imports 1000 employees with roles

Role Inheritance and Access Policies

Role Inheritance refers to the automatic assignment of roles (and their associated permissions) to users based on their membership in specific groups or divisions.
For example:

If a user is added to the "Pharmacy" group, they inherit the "Pharmacy Manager" role automatically, along with permissions like "View Patient Records" or "Process Orders."

This eliminates manual assignments and ensures consistent access control across teams.
Access Policies are rules that govern how roles and permissions are assigned, enforced, and managed. These policies define:

1. Inheritance Rules: Whether roles are auto assigned via group membership (for new applications) or manually assigned (for legacy systems).

2. Conditions: Criteria for granting/revoking access (e.g., job function, location).

3. Compliance: Requirements like periodic access reviews or separation of duties.

4. Custom Attributes: Metadata (e.g., "Department=Cardiology") used to refine access decisions.

- Purpose: Define custom attributes for users and groups to aid in policy enforcement and decision-making.
- Steps:
    1. Navigate to the **Custom Attributes** section.
    2. Define new custom attributes as required.
    3. Assign custom attributes to users and groups

Example:

A hospital's "Surgery" group has an access policy that auto-assigns the "Surgeon" role to all members. This role includes permissions like "Access Operating Theater" and "Update Patient Charts."

If a user is removed from the group, the role (and permissions) is automatically revoked, ensuring compliance with least privilege principles.

In essence, role inheritance streamlines access management, while access policies enforce security, compliance, and operational efficiency.

Access Governance Framework

The Access Governance Framework in the Admin Console IGA solution is designed to enforce secure, scalable, and self-service access management across applications and user roles. Below is a structured breakdown

1. Core Components of the Framework

Component	Description
Persona-Based Access Control	Supports five key personas: Super Admin, Application Admin, Group Owner, User Admin, and Org Owner. Each has defined privileges (e.g., Super Admins oversee everything; Group Owners manage division-specific access).
Role & Permission Modeling	Roles (e.g., "Pharmacy Manager") are created with attached permissions (actions like "view" or "edit"). Permissions are grouped into roles for streamlined assignment.
Group-Based Access	Users inherit roles via group membership (e.g., "Pharmacy" group members auto-get "Pharmacy_Admin" role). Supports **auto-inheritance** (new apps) or **manual assignment** (legacy apps).
Custom Attributes	Additional metadata (e.g., "Department=Cardiology") can be assigned to users/groups for fine-grained policy enforcement.

2. Governance Mechanisms

Mechanism	How It Works	Purpose
Self-Service Access Assignment	Admins/owners assign roles via UI or bulk CSV. Users can be invited with pre-assigned access (Fast-Track Registration).	Reduces dependency on helpdesk; accelerates onboarding.
Auto-Revocation	Removing a user from a group revokes all associated roles instantly.	Ensures compliance and reduces "orphaned" access.
Bulk Import/Export	CSV files assign/revoke access for thousands of users (e.g., onboarding new hires).	Scalable for enterprise use.
Application Lifecycle Control	Super Admins can deactivate applications, blocking all authentication attempts.	Mitigates risks from compromised portals.

Note:
Governance Principles

- **Least Privilege**: Roles/permissions are scoped to minimal necessary access.
- **Separation of Duties**: Personas restrict visibility/actions (e.g., Group Owners only see their groups).
- **Auditability**: All access changes are logged; future features will include reporting/analytics

Compliance and Access Policies

Policy-Based Role Management

Objective: Streamline role assignments using structured policies.

1. Role Creation

- Navigate to **Applications** > **Roles in Identity Console**.
- Define roles (e.g., "User Administrator," "Application Owner") with descriptions.

2. Permission Assignment

- Under each role, attach permissions (e.g., "Manage Users," "Configure SSO").
- Permissions are predefined in the system (e.g., OAuth scopes, API access).

3. Role-to-Group Mapping

- Assign roles to groups (e.g., "Pharmacy" group gets "Pharmacy Manager" role).
- For migrated applications, manually assign roles via **Users** > **Access** > **Assign Role**.

4. Hierarchical Access Control

- Super Admin: Manages all applications and users.
- Application Owner: Manages roles/groups within their assigned application.
- Group Owner: Manages users and roles within their group (e.g., "Pharmacy" group).

Audit and Reporting Features

Objectives:

- Provides visibility into user access.
- Enables compliance audits.
- If a user is part of a group, they automatically inherit associated permissions.
- Direct role assignment is discouraged to maintain better governance.

1. Audit Logs

    - User access grants/revocations.
    - Role/group modifications.
    - Application lifecycle changes (policy enforce activation/deactivation).

2. Reporting
    - User Access Summary: List roles/permissions per user.
    - Group Compliance: Highlight groups with outdated roles.
    - Application Status: Track active/inactive applications.

3. Periodic Recertification
    - Schedule reviews for role assignments and group memberships.
    - Flag orphaned roles or overprivileged users.

4. Group-Based Access Control (GBAC)
   Objective: Simplify access management through groups.

    - Access is controlled through groups rather than direct assignments.
    - Reduces governance complexity and ensures compliance.
    - Group-based policies ensure roles are inherited. New applications have an auto-inheritance policy for roles, while migrated applications require manual role assignment.
    - Reduces manual assignment errors.

Steps

1. Group Creation

    a. Navigate to **Applications** > **Groups**.

b. Define groups (e.g., "OPD," "Cardiology") with descriptions and custom attributes.

2. Role Inheritance

a. Assign roles to groups (e.g., "OPD" group gets "View Patient Records" role).
b. Users added to the group inherit roles automatically (if auto-inheritance is enabled).

3. User Assignment

Add users via:

- Bulk Import: Upload CSV with user emails and target groups.
- Manual Entry: Search users in the tenancy and assign to groups.

4. Access Revocation

- Remove users from groups to revoke inherited roles.
- For manual assignments, revoke roles individually under Users > Access.

5. Self-Service Management

- Group Owners: Can add/remove users and edit group roles without Super Admin intervention.
- Application Owners: Oversee all groups under their application and enforce policies

Access Revocation and Governance

Access revocation is the process of removing or restricting a user's access to applications, roles and permissions within an organization.

Objectives

- When a user leaves or changes roles, access is revoked automatically by removing them from groups.
- Governance ensures that only necessary access is granted

Method 1: Group-Based Access Revocation
Group Owner or Super Admin is responsible for carrying out this action.

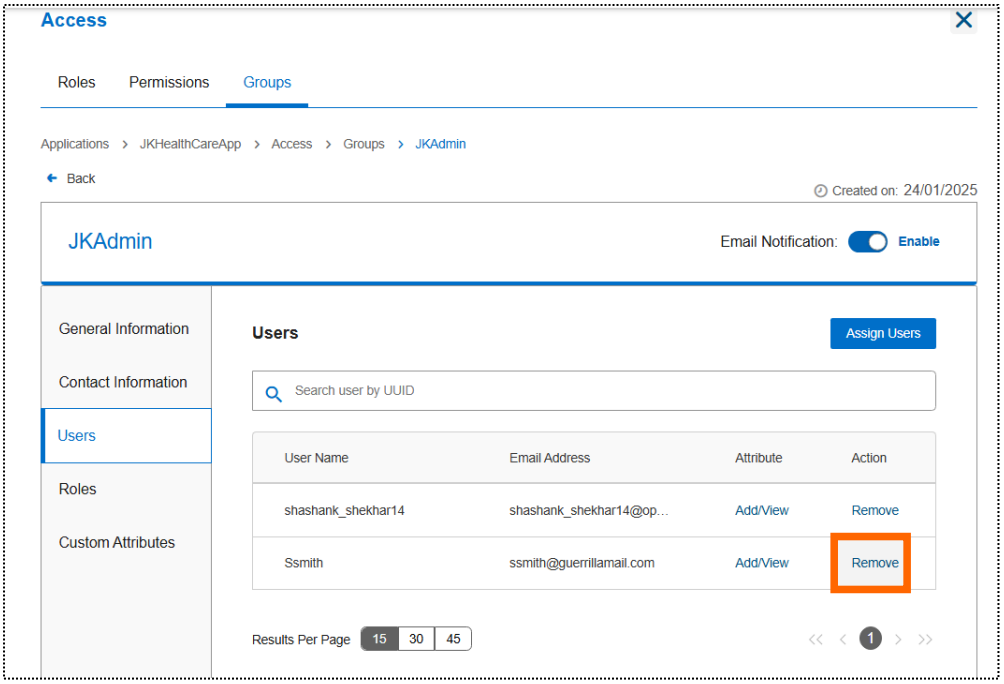
Steps:

1. **Applications** > [**Select Application**] > **Access** > **Groups** > [**Select Group**].
   Example: For the JKAdmin group under JK Healthcare application.

2. Locate User in the Group:
   Under the group's **Users** tab, search for the target user (e.g., Steve Smith).

3. Revoke Group Membership:
   Click **Remove** next to the user's name.
   All roles auto inherited via the group are revoked immediately.

Example: Revoking Steve Smith from the **JKAdmin** group removes his **Doctor** role.



Method 2: Manual Role Revocation
Application Owner or Super Admin are responsible for carrying out this action.

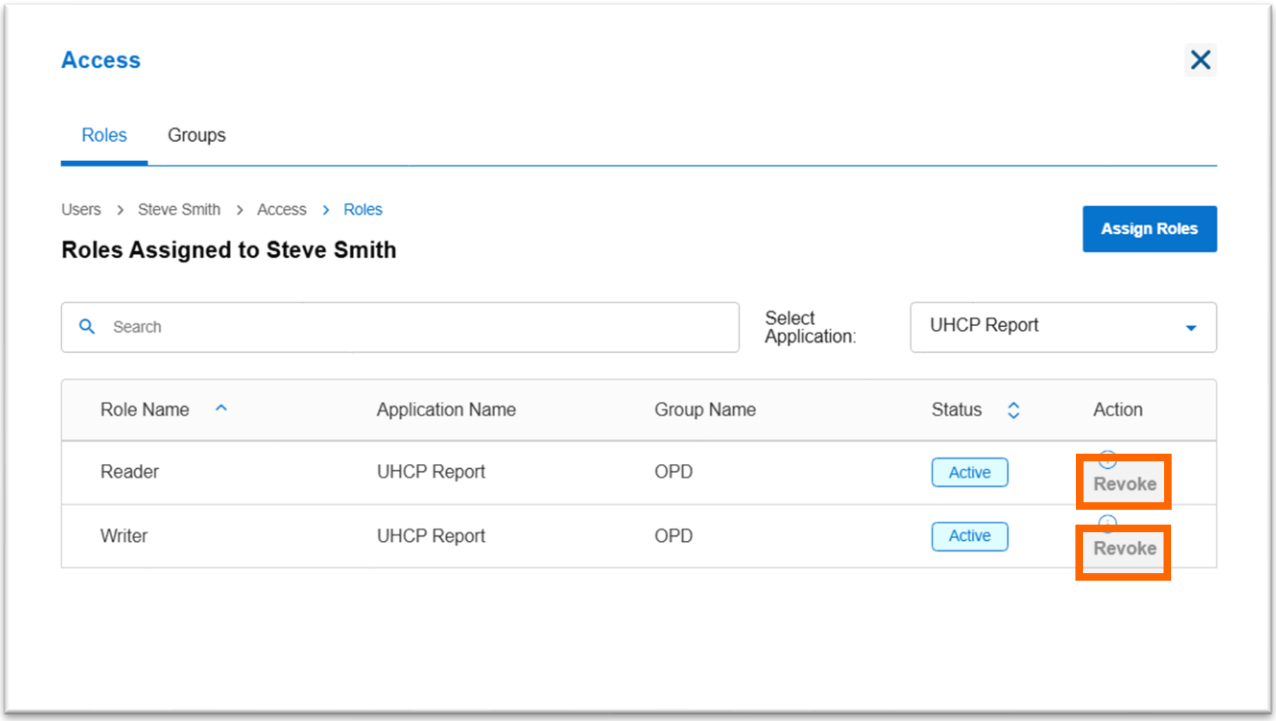
For legacy applications with no auto-inheritance.

Steps:

1. **Users** > **Search User** > Select [**username**]> **Access**.

2. Search for User and Application:
   Enter the username (e.g., "Steve Smith") and select the application (e.g., "UHCP Report").

3. Revoke Assigned Roles:
   Under the Roles section, click **Revoke** next to the role to remove.
   Directly assigned roles are revoked, but group memberships remain intact.



Governance Policies

Governance policies in the Admin Console's IGA system are rule-based frameworks that enforce secure, compliant, and standardized access controls across an organization. These policies automate decision-making for access provisioning, revocation, and role assignments while ensuring alignment with regulatory requirements (e.g., HIPAA, SOX).

Objective: Enforce compliance through structured access control and policies.

Policy 1: Auto-Inheritance for New Applications
Super Admins, Application owners and Organization owners are responsible to perform this action
Steps:

1. **Applications** > [**New Application**] > **Settings** > **Group Policy**.
2. Toggle Auto-Inheritance to **On**.
   Result: Users added to groups automatically inherit roles assigned to those groups.
3. Assign Roles to Groups: Attach roles (e.g., "Surgeon") to groups (e.g., "Surgery") under **Groups** > **Roles**.

Policy 2: Custom Attributes for Enhanced Control
Application owners and Organization owners are responsible to perform this action.
Steps:

1. **Applications** > [**Application**] > **Custom Attributes**
   Add metadata (e.g., Department=Cardiology, Location=NYC).
2. Apply Attributes to Users/Groups:
   Use attributes to enforce policies (e.g., restrict access to users with Location=NYC).

Policy 3: Application Lifecycle Management
Super Admin or Application Owner can manage the Application Status.
Steps:

1. **Applications** > [**Name of the Application**] > **Status**.
   Set to **Inactive** to block all authentication requests (e.g., during a security breach).

User Access Management

User Access Management is the end-to-end process of provisioning, modifying, and revoking user access to applications and resources within the IGA (Identity Governance and Administration) framework. It combines self-service capabilities and role-based controls to ensure secure and efficient access lifecycle management.

Core Components of User Access Management

- **Personas & Roles**
  - **Five Key Personas**:
    - **Super Admin**: Full control over all applications/users.

    - **Application Admin**: Manages specific applications.
    - **Group Owner**: Manages access for a division (e.g., "Pharmacy" group).
    - **User Admin**: Handles user lifecycle (onboarding/offboarding).
    - **Org Owner**: Oversees tenant-level policies.

- **Roles & Permissions**:
    - Roles (e.g., "Pharmacy_Admin") bundle permissions (e.g., "view records", "edit orders").
    - Assigned manually or inherited via group membership.

b. Access Assignment Methods

Method	Process	Use Case
Manual Assignment	Admins assign roles to users via UI.	Ad-hoc access needs.
Bulk Import (CSV)	Upload predefined roles/groups for mass onboarding.	Large-scale hires (e.g., 10K users).
Fast-Track Registration	Invite users with pre-assigned roles; access granted upon registration.	Critical roles (e.g., healthcare providers).
Group Inheritance	Users auto-inherit roles when added to a group.	Team/department-based access (e.g., "Finance" group).

c. Access Revocation

- **Group-Level**: Remove user → All group roles revoked instantly.
- **Role-Level**: Selective revocation of individual roles.
- **Bulk Revocation**: Revoke all roles for a user/group via CSV.

Use Cases

Hospital Management System

- Doctors, nurses, and billing staff have separate access levels.
- Only authorized users can enter restricted areas like the operation theater.

Enterprise Application Access

- Employees are assigned to departments with predefined access.
- Access is revoked automatically when an employee leaves.

Healthcare Provider Onboarding

- New providers receive access on day one.
- They can immediately access necessary applications without delays.